

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		30339	
Total Threats Mitigated	47	Total Packets Analyzed		1927920770	
High Priority	16	Malicious Packets		1921868930	
Low Priority	31	True Negatives Passed		24978	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		10	
ML Metrics	Value	SYN Flood		15	
False Positives	44	UDP Flood		9	
FP Rate	0.18%	Uncertain		13	
IF Precision	99.06%				
IF Recall	97.31%	Action		Count	
IF F1-Score	98.18%	Quarantined		10	
IF Accuracy	99.42%	Time Ban		10	
		Blackhole		0	
		Released		10	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.06%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.31%	Share of actual attacks successfully flagged
F1-Score	98.18%	Balanced measure combining Precision and Recall
Accuracy	99.42%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.18%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.69%	Actual attacks that went undetected
True Positive Rate (TPR)	97.31%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.82%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4635	FN 128
Actual: Normal	FP 44	TN 24850

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	2711	0	0
Act: ICMP	0	581	0
Act: UDP	0	0	1343

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.06%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.31%	Share of actual attacks successfully flagged
F1-Score	98.18%	Balanced measure combining Precision and Recall
Accuracy	99.42%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.18%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.69%	Actual attacks that went undetected
True Positive Rate (TPR)	97.31%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.82%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4635	FN 128
Actual: Normal	FP 44	TN 24850

2d. Response Latency

Metric	Value	Description
Detection Time	509.2 ms	Average time taken to detect an attack
Mitigation Response Time	30.7 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	2.77%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	15.35%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	6.76%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.25	6	5	2026-09-25 14:29:27	2026-09-25 14:35:29
10.0.0.18	5	5	2026-09-25 14:29:27	2026-09-25 14:33:38
10.0.0.16	3	3	2026-09-25 14:32:51	2026-09-25 14:35:23
10.0.0.17	3	3	2026-09-25 14:32:51	2026-09-25 14:35:25
10.0.0.22	3	2	2026-09-25 14:33:14	2026-09-25 14:35:36
10.0.0.19	4	1	2026-09-25 14:29:27	2026-09-25 14:32:54
10.0.0.20	2	1	2026-09-25 14:33:22	2026-09-25 14:35:24
10.0.0.21	2	1	2026-09-25 14:33:22	2026-09-25 14:35:30

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.1	1	1	2026-09-25 14:33:01	2026-09-25 14:33:12
10.0.0.23	1	1	2026-09-25 14:35:15	2026-09-25 14:35:26
10.0.0.24	1	1	2026-09-25 14:35:17	2026-09-25 14:35:27
10.0.0.3	1	1	2026-09-25 14:33:03	2026-09-25 14:33:13

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:29:26	10.0.0.21	DDoS	UDP Flood	86.5%	Critical	Quarantined
2026-09-25 14:29:27	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Quarantined
2026-09-25 14:29:27	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Quarantined
2026-09-25 14:29:27	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Quarantined
2026-09-25 14:29:37	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban (30s)
2026-09-25 14:29:37	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (30s)
2026-09-25 14:29:37	10.0.0.19	DDoS	SYN Flood	93.4%	High	Released
2026-09-25 14:30:37	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban (1m)
2026-09-25 14:30:37	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (1m)
2026-09-25 14:31:37	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban (2m)
2026-09-25 14:31:37	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-25 14:32:45	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban
2026-09-25 14:32:51	10.0.0.17	DDoS	SYN Flood	95.9%	High	Time Ban (30s)
2026-09-25 14:32:51	10.0.0.16	DDoS	SYN Flood	90.4%	High	Time Ban (30s)
2026-09-25 14:32:51	10.0.0.17	DDoS	SYN Flood	95.9%	High	Time Ban
2026-09-25 14:32:51	10.0.0.16	DDoS	SYN Flood	90.4%	High	Time Ban
2026-09-25 14:32:54	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-25 14:32:55	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban
2026-09-25 14:33:12	10.0.0.1	DDoS	Uncertain	60.6%	Low	Released
2026-09-25 14:33:13	10.0.0.3	DDoS	Uncertain	58.8%	Low	Released
2026-09-25 14:33:14	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:33:22	10.0.0.20	DDoS	UDP Flood	78.1%	High	Time Ban (30s)
2026-09-25 14:33:22	10.0.0.20	DDoS	UDP Flood	78.1%	High	Time Ban
2026-09-25 14:33:22	10.0.0.21	DDoS	UDP Flood	86.5%	High	Time Ban (30s)
2026-09-25 14:33:22	10.0.0.21	DDoS	UDP Flood	86.5%	High	Time Ban
2026-09-25 14:33:25	10.0.0.22	DDoS	UDP Flood	82.0%	High	Time Ban (30s)
2026-09-25 14:33:25	10.0.0.22	DDoS	UDP Flood	82.0%	Critical	Time Ban
2026-09-25 14:33:38	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban (5m)
2026-09-25 14:33:38	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 14:33:45	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-25 14:33:51	10.0.0.23	DDoS	ICMP Flood	95.8%	High	Time Ban (30s)
2026-09-25 14:33:51	10.0.0.23	DDoS	ICMP Flood	95.8%	High	Time Ban
2026-09-25 14:33:51	10.0.0.24	DDoS	ICMP Flood	99.6%	High	Time Ban (30s)
2026-09-25 14:33:51	10.0.0.24	DDoS	ICMP Flood	99.6%	High	Time Ban
2026-09-25 14:33:52	10.0.0.20	DDoS	UDP Flood	82.7%	High	Released
2026-09-25 14:33:52	10.0.0.21	DDoS	UDP Flood	86.5%	High	Released
2026-09-25 14:35:13	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:35:13	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:35:15	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:35:15	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:35:17	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:35:23	10.0.0.16	DDoS	SYN Flood	93.2%	Critical	Time Ban (1m)
2026-09-25 14:35:25	10.0.0.17	DDoS	Uncertain	-	Low	Released
2026-09-25 14:35:26	10.0.0.23	DDoS	Uncertain	-	Low	Released
2026-09-25 14:35:27	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-25 14:35:29	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-09-25 14:35:36	10.0.0.22	DDoS	Uncertain	-	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:35:36	Attack Stopped
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:35:30	Attack Stopped
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:35:29	Attack Stopped
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:35:27	Attack Stopped
10.0.0.23	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:35:26	Attack Stopped
10.0.0.17	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:35:25	Attack Stopped
10.0.0.20	UDP Flood	0.6294	86.3%	Critical	Phase 2	1	11s	2026-09-25 14:35:24	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6521	93.2%	Critical	Phase 2	1	10s	2026-09-25 14:35:23	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6105	85.9%	High	Phase 2	2	41s	2026-09-25 14:33:55	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6472	86.5%	High	Phase 2	1	30s	2026-09-25 14:33:52	Ban Expired
10.0.0.20	UDP Flood	0.6358	82.7%	High	Phase 2	1	30s	2026-09-25 14:33:52	Ban Expired
10.0.0.16	SYN Flood	0.6512	93.3%	Critical	Phase 2	3	1m 1s	2026-09-25 14:33:52	Escalated to Time Ban
10.0.0.17	SYN Flood	0.6199	95.9%	Critical	Phase 2	3	1m 1s	2026-09-25 14:33:52	Escalated to Time Ban
10.0.0.18	SYN Flood	0.6370	93.4%	Critical	Phase 2	5	4m 11s	2026-09-25 14:33:38	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6461	91.2%	Critical	Phase 2	5	4m 11s	2026-09-25 14:33:38	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6398	82.0%	High	Phase 2	1	11s	2026-09-25 14:33:25	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6128	93.3%	High	Phase 2	2	31s	2026-09-25 14:33:22	Escalated to Time Ban
10.0.0.17	SYN Flood	0.6048	95.9%	High	Phase 2	2	31s	2026-09-25 14:33:22	Escalated to Time Ban
10.0.0.3	Uncertain	0.6126	58.8%	Low	Phase 1	1	10s	2026-09-25 14:33:13	Attack Stopped
10.0.0.1	Uncertain	0.5931	60.6%	Low	Phase 1	1	11s	2026-09-25 14:33:12	Attack Stopped
10.0.0.19	SYN Flood	0.6516	93.4%	Critical	Phase 2	1	10s	2026-09-25 14:32:54	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6319	93.4%	Critical	Phase 2	4	2m 10s	2026-09-25 14:31:37	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6461	91.2%	Critical	Phase 2	4	2m 10s	2026-09-25 14:31:37	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6319	93.4%	Critical	Phase 2	3	1m 10s	2026-09-25 14:30:37	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6461	91.2%	Critical	Phase 2	3	1m 10s	2026-09-25 14:30:37	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6319	93.4%	Critical	Phase 2	2	40s	2026-09-25 14:30:07	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6461	91.2%	Critical	Phase 2	2	40s	2026-09-25 14:30:07	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6309	93.4%	Critical	Phase 1	1	11s	2026-09-25 14:30:01	Attack Stopped
10.0.0.19	SYN Flood	0.6253	93.4%	Critical	Phase 1	1	11s	2026-09-25 14:29:49	Attack Stopped
10.0.0.19	SYN Flood	0.6222	93.4%	High	Phase 1	1	10s	2026-09-25 14:29:37	Attack Stopped
10.0.0.18	SYN Flood	0.6319	93.4%	Critical	Phase 2	1	10s	2026-09-25 14:29:37	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6461	91.2%	Critical	Phase 2	1	10s	2026-09-25 14:29:37	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE